Data Analytics (DA)

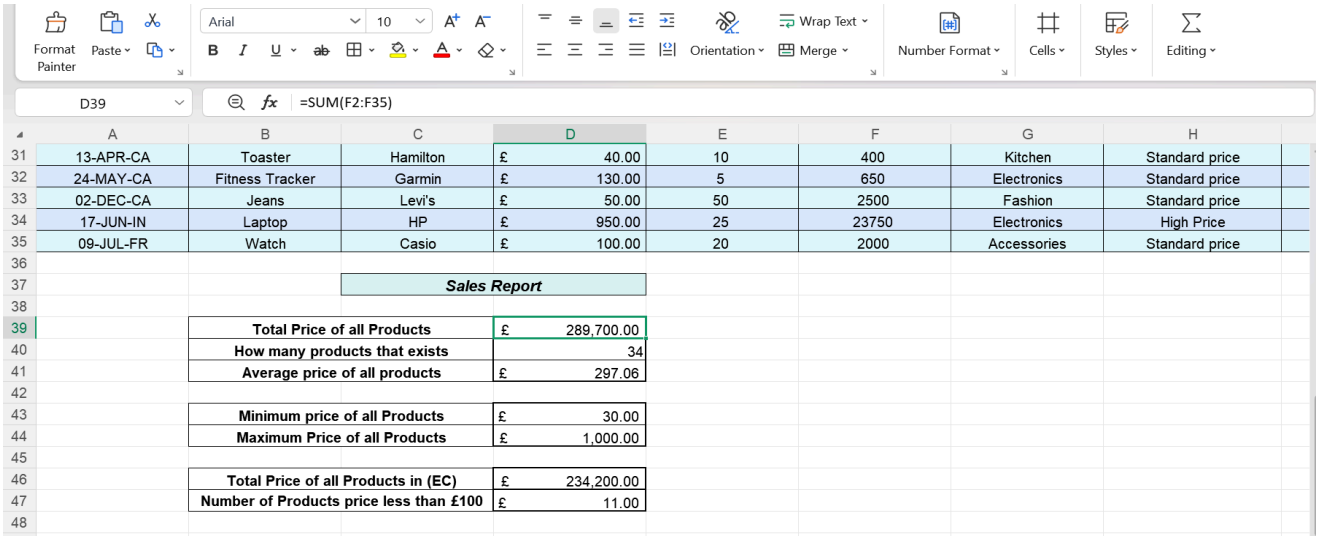
Module -1 - Excel

Excel Assignment 1 – Data Exploration

Basic Data Exploration

Step 1: Sum, Count, and Average

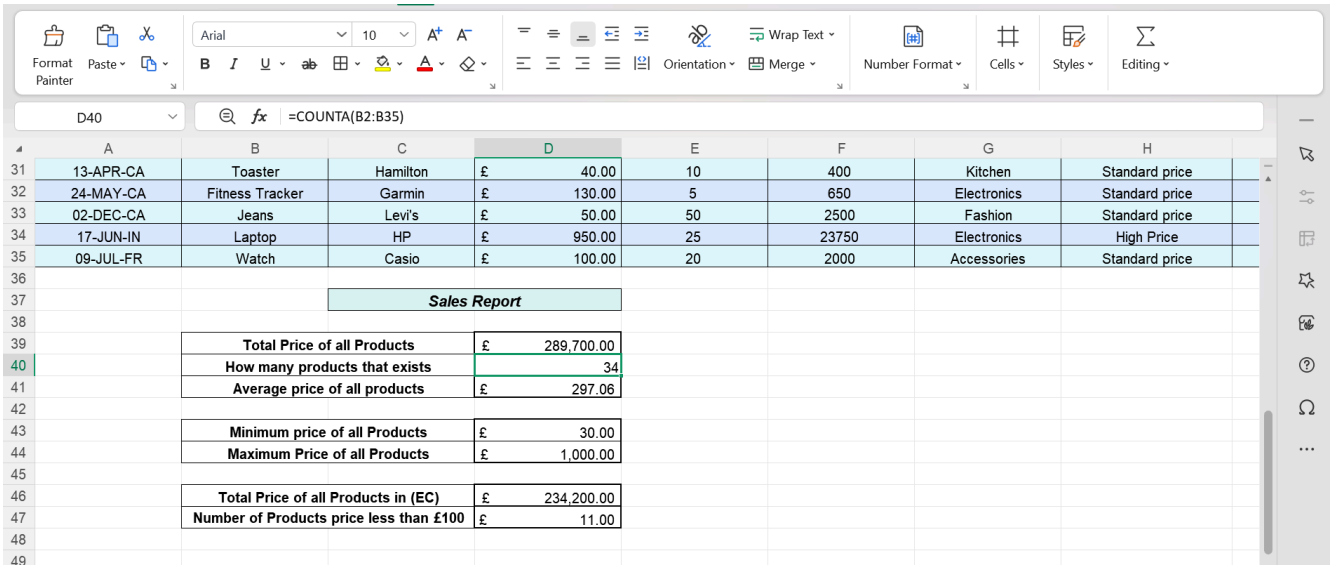
Total price of all products: (SUM).



	A	B	C	D	E	F	G	H
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	400	Kitchen	Standard price
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	650	Electronics	Standard price
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	2500	Fashion	Standard price
34	17-JUN-IN	Laptop	HP	£ 950.00	25	23750	Electronics	High Price
35	09-JUL-FR	Watch	Casio	£ 100.00	20	2000	Accessories	Standard price

Sales Report	
Total Price of all Products	£ 289,700.00
How many products that exists	34
Average price of all products	£ 297.06
Minimum price of all Products	£ 30.00
Maximum Price of all Products	£ 1,000.00
Total Price of all Products in (EC)	£ 234,200.00
Number of Products price less than £100	£ 11.00

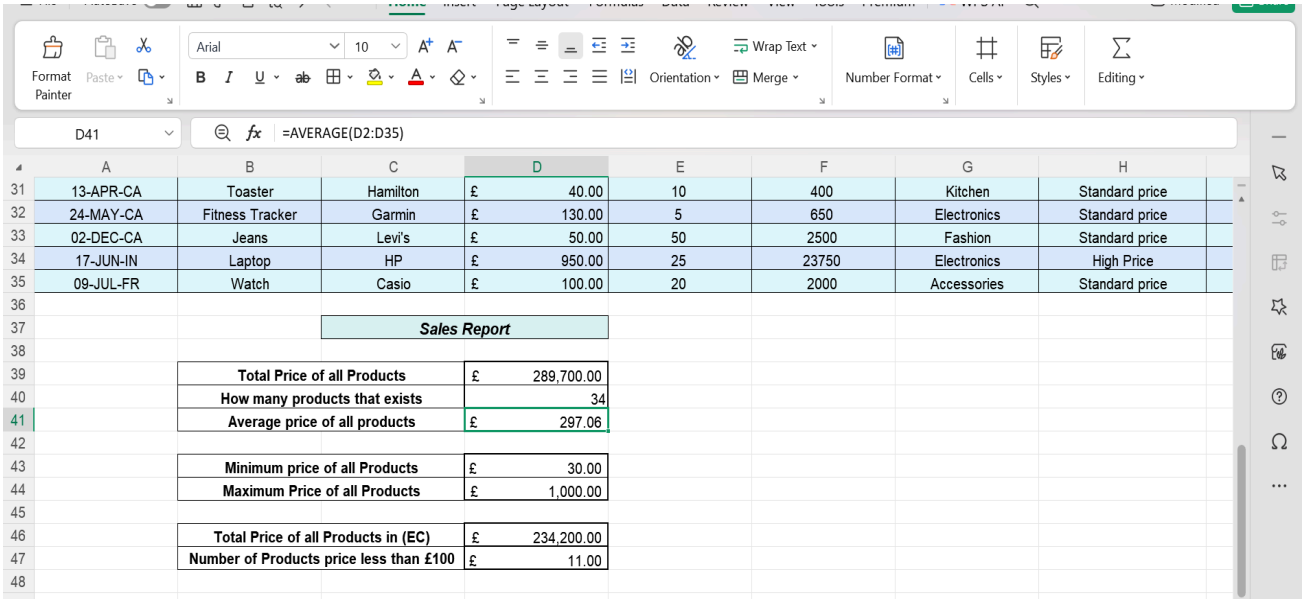
How many products exist in the dataset: (COUNTA).



	A	B	C	D	E	F	G	H
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	400	Kitchen	Standard price
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	650	Electronics	Standard price
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	2500	Fashion	Standard price
34	17-JUN-IN	Laptop	HP	£ 950.00	25	23750	Electronics	High Price
35	09-JUL-FR	Watch	Casio	£ 100.00	20	2000	Accessories	Standard price

Sales Report	
Total Price of all Products	£ 289,700.00
How many products that exists	34
Average price of all products	£ 297.06
Minimum price of all Products	£ 30.00
Maximum Price of all Products	£ 1,000.00
Total Price of all Products in (EC)	£ 234,200.00
Number of Products price less than £100	£ 11.00

Average price of the products: (AVERAGE).

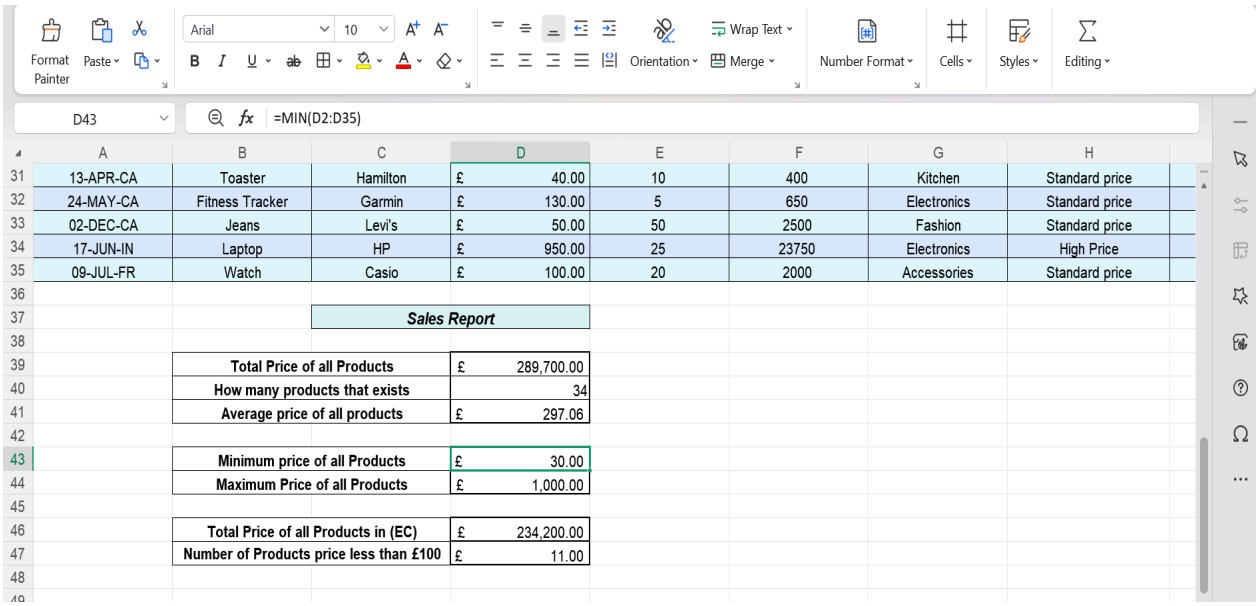


D41 =AVERAGE(D2:D35)

	A	B	C	D	E	F	G	H
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	400	Kitchen	Standard price
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	650	Electronics	Standard price
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	2500	Fashion	Standard price
34	17-JUN-IN	Laptop	HP	£ 950.00	25	23750	Electronics	High Price
35	09-JUL-FR	Watch	Casio	£ 100.00	20	2000	Accessories	Standard price
36								
37			***Sales Report***					
38								
39		**Total Price of all Products**		£ 289,700.00				
40		**How many products that exists**		34				
41		**Average price of all products**		£ 297.06				
42								
43		**Minimum price of all Products**		£ 30.00				
44		**Maximum Price of all Products**		£ 1,000.00				
45								
46		**Total Price of all Products in (EC)**		£ 234,200.00				
47		**Number of Products price less than £100**		£ 11.00				
48								

Step 2: Minimum and Maximum Values

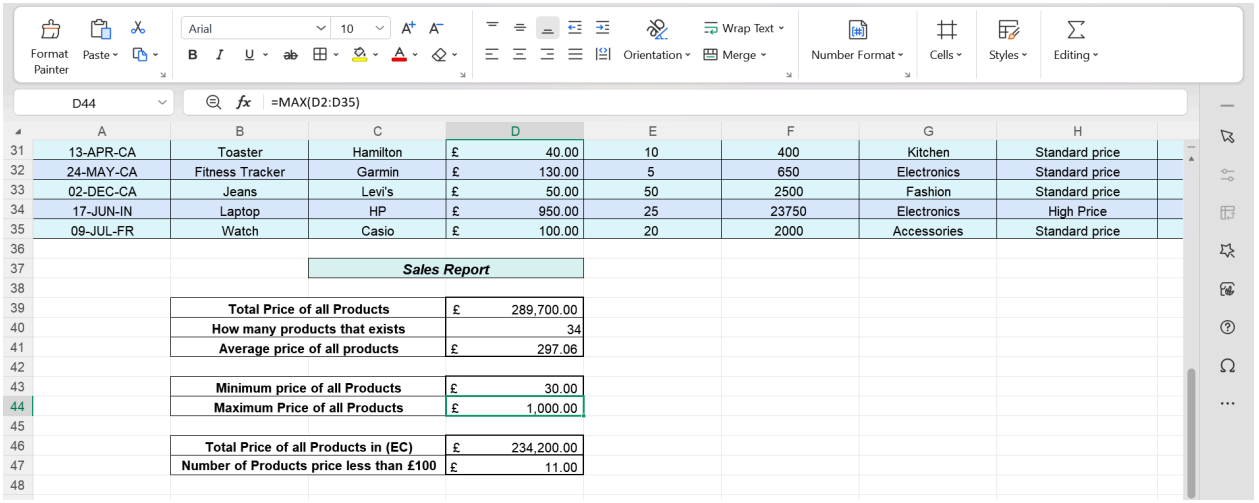
Minimum price: (MIN).



D43 =MIN(D2:D35)

	A	B	C	D	E	F	G	H
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	400	Kitchen	Standard price
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	650	Electronics	Standard price
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	2500	Fashion	Standard price
34	17-JUN-IN	Laptop	HP	£ 950.00	25	23750	Electronics	High Price
35	09-JUL-FR	Watch	Casio	£ 100.00	20	2000	Accessories	Standard price
36								
37			***Sales Report***					
38								
39		**Total Price of all Products**		£ 289,700.00				
40		**How many products that exists**		34				
41		**Average price of all products**		£ 297.06				
42								
43		**Minimum price of all Products**		£ 30.00				
44		**Maximum Price of all Products**		£ 1,000.00				
45								
46		**Total Price of all Products in (EC)**		£ 234,200.00				
47		**Number of Products price less than £100**		£ 11.00				
48								

Maximum price: (MAX).



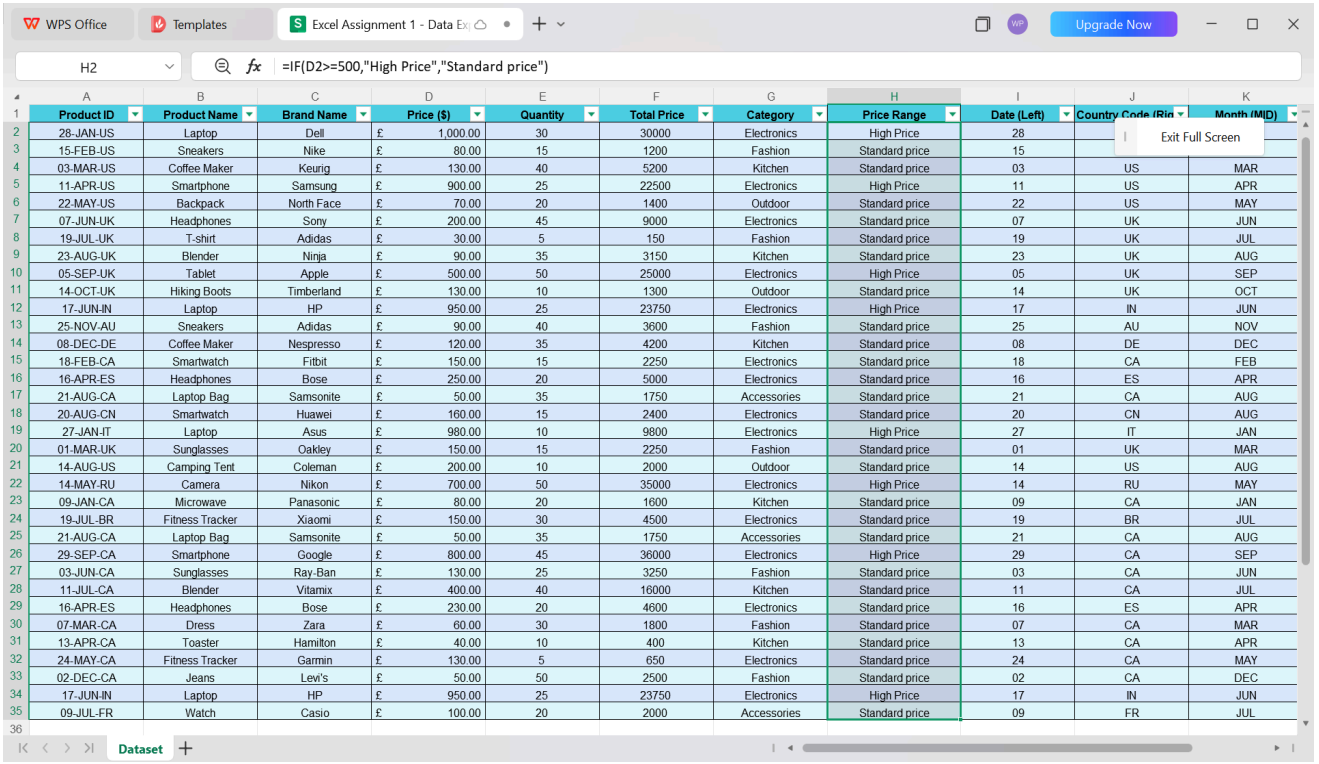
	A	B	C	D	E	F	G	H
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	400	Kitchen	Standard price
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	650	Electronics	Standard price
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	2500	Fashion	Standard price
34	17-JUN-IN	Laptop	HP	£ 950.00	25	23750	Electronics	High Price
35	09-JUL-FR	Watch	Casio	£ 100.00	20	2000	Accessories	Standard price

Sales Report	
Total Price of all Products	£ 289,700.00
How many products that exists	34
Average price of all products	£ 297.06
Minimum price of all Products	£ 30.00
Maximum Price of all Products	£ 1,000.00
Total Price of all Products in (EC)	£ 234,200.00
Number of Products price less than £100	£ 11.00

Step 3: Logical Function – IF Condition

If Price ≥ $500 → classify as (High Price).

If Price < $500 → classify as (Standard Price).

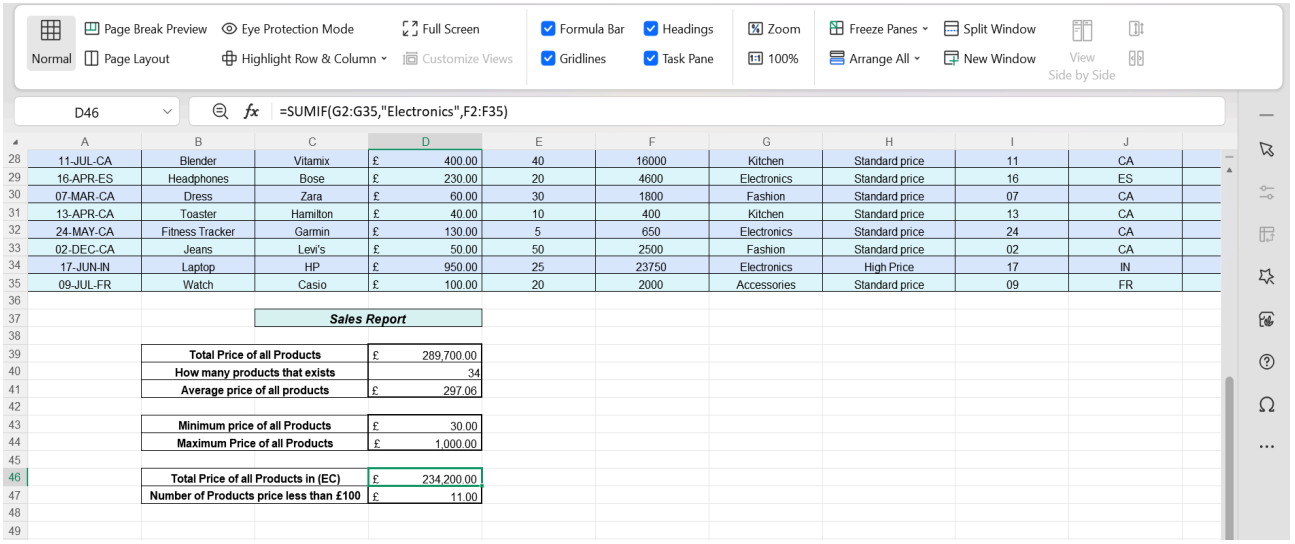


Product ID	Product Name	Brand Name	Price ($)	Quantity	Total Price	Category	Price Range	Date (Left)	Country Code (Ric	Month (MID)
28-JAN-US	Laptop	Dell	£ 1,000.00	30	30000	Electronics	High Price	28		
15-FEB-US	Sneakers	Nike	£ 80.00	15	1200	Fashion	Standard price	15		
03-MAR-US	Coffee Maker	Keurig	£ 130.00	40	5200	Kitchen	Standard price	03	US	MAR
11-APR-US	Smartphone	Samsung	£ 900.00	25	22500	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	£ 70.00	20	1400	Outdoor	Standard price	22	US	MAY
07-JUN-UK	Headphones	Sony	£ 200.00	45	9000	Electronics	Standard price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	£ 30.00	5	150	Fashion	Standard price	19	UK	JUL
23-AUG-UK	Blender	Ninja	£ 90.00	35	3150	Kitchen	Standard price	23	UK	AUG
05-SEP-UK	Tablet	Apple	£ 500.00	50	25000	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	£ 130.00	10	1300	Outdoor	Standard price	14	UK	OCT
17-JUN-IN	Laptop	HP	£ 950.00	25	23750	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	£ 90.00	40	3600	Fashion	Standard price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	£ 120.00	35	4200	Kitchen	Standard price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	£ 150.00	15	2250	Electronics	Standard price	18	CA	FEB
16-APR-ES	Headphones	Bose	£ 250.00	20	5000	Electronics	Standard price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	£ 50.00	35	1750	Accessories	Standard price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	£ 160.00	15	2400	Electronics	Standard price	20	CN	AUG
27-JAN-IT	Laptop	Asus	£ 980.00	10	9800	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	£ 150.00	15	2250	Fashion	Standard price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	£ 200.00	10	2000	Outdoor	Standard price	14	US	AUG
14-MAY-RU	Camera	Nikon	£ 700.00	50	35000	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	£ 80.00	20	1600	Kitchen	Standard price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	£ 150.00	30	4500	Electronics	Standard price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	£ 50.00	35	1750	Accessories	Standard price	21	CA	AUG
29-SEP-CA	Smartphone	Google	£ 800.00	45	36000	Electronics	High Price	29	CA	SEP
03-JUN-CA	Sunglasses	Ray-Ban	£ 130.00	25	3250	Fashion	Standard price	03	CA	JUN
11-JUL-CA	Blender	Vitamix	£ 400.00	40	16000	Kitchen	Standard price	11	CA	JUL
16-APR-ES	Headphones	Bose	£ 230.00	20	4600	Electronics	Standard price	16	ES	APR
07-MAR-CA	Dress	Zara	£ 60.00	30	1800	Fashion	Standard price	07	CA	MAR
13-APR-CA	Toaster	Hamilton	£ 40.00	10	400	Kitchen	Standard price	13	CA	APR
24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	650	Electronics	Standard price	24	CA	MAY
02-DEC-CA	Jeans	Levi's	£ 50.00	50	2500	Fashion	Standard price	02	CA	DEC
17-JUN-IN	Laptop	HP	£ 950.00	25	23750	Electronics	High Price	17	IN	JUN
09-JUL-FR	Watch	Casio	£ 100.00	20	2000	Accessories	Standard price	09	FR	JUL

Step 4: Conditional Functions

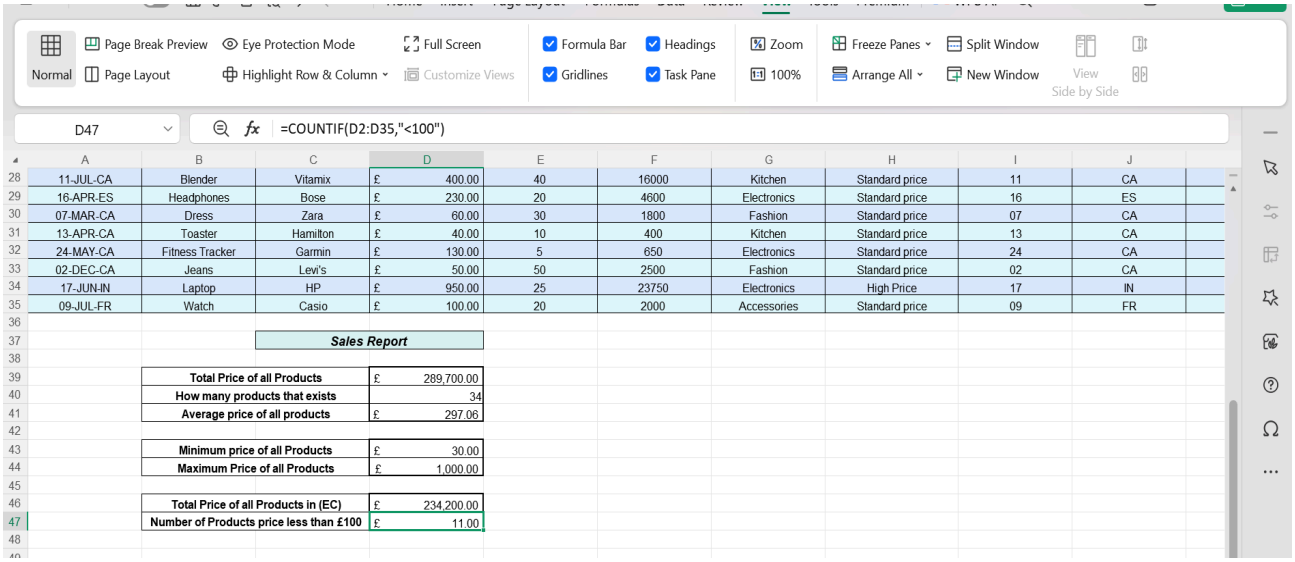
- **SUMIF**
- **COUNTIF**

Total price of products in the Electronics category - (SUMIF).



	A	B	C	D	E	F	G	H	I	J
28	11-JUL-CA	Blender	Vitamix	£ 400.00	40	16000	Kitchen	Standard price	11	CA
29	16-APR-ES	Headphones	Bose	£ 230.00	20	4600	Electronics	Standard price	16	ES
30	07-MAR-CA	Dress	Zara	£ 60.00	30	1800	Fashion	Standard price	07	CA
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	400	Kitchen	Standard price	13	CA
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	650	Electronics	Standard price	24	CA
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	2500	Fashion	Standard price	02	CA
34	17-JUN-IN	Laptop	HP	£ 950.00	25	23750	Electronics	High Price	17	IN
35	09-JUL-FR	Watch	Casio	£ 100.00	20	2000	Accessories	Standard price	09	FR
36										
37			***Sales Report***							
38										
39		**Total Price of all Products**		£ 289,700.00						
40		**How many products that exists**		34						
41		**Average price of all products**		£ 297.06						
42										
43		**Minimum price of all Products**		£ 30.00						
44		**Maximum Price of all Products**		£ 1,000.00						
45										
46		**Total Price of all Products in (EC)**		£ 234,200.00						
47		**Number of Products price less than £100**		£ 11.00						
48										
49										

Number of products with a price less than $100 - (COUNTIF)



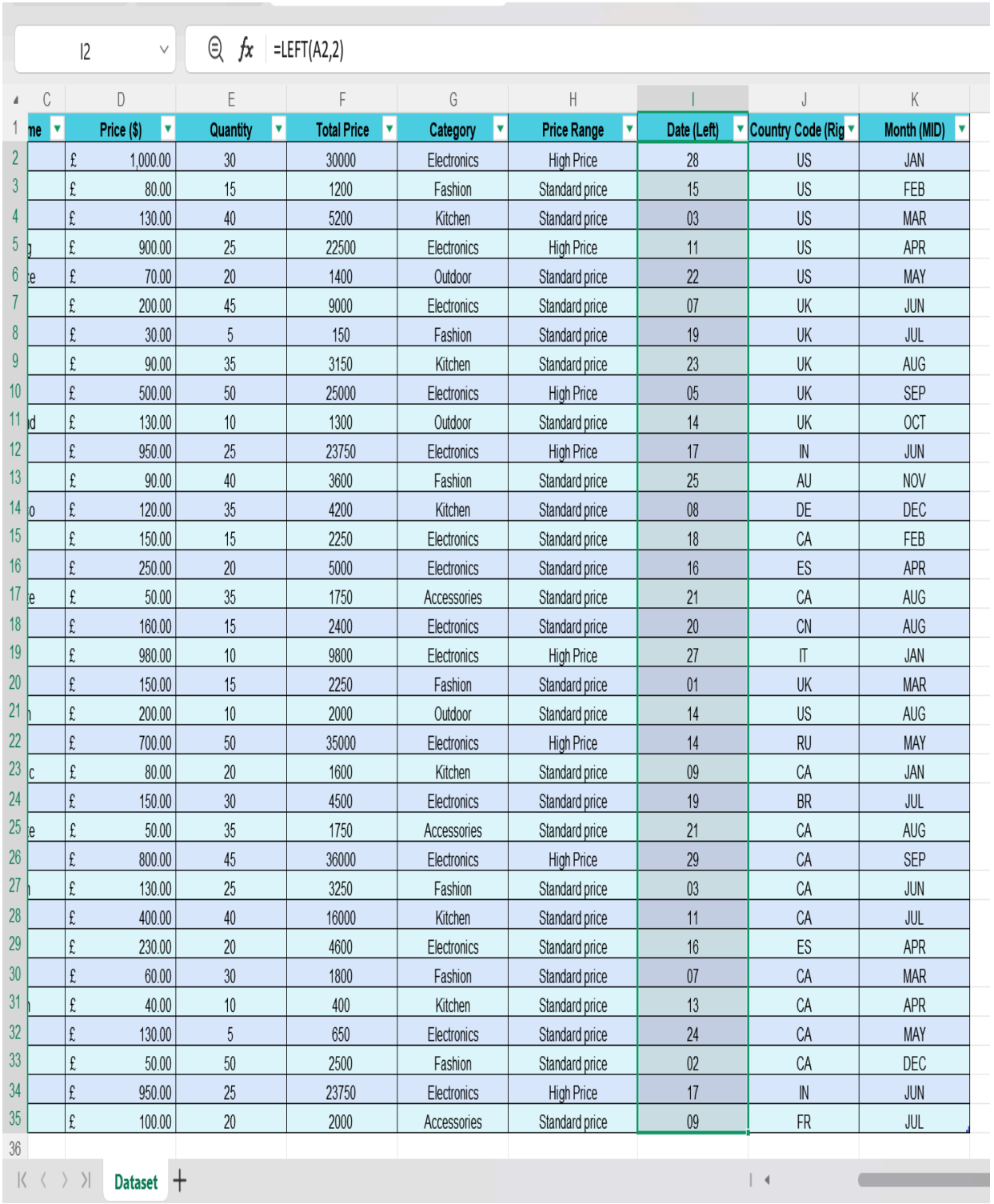
	A	B	C	D	E	F	G	H	I	J
28	11-JUL-CA	Blender	Vitamix	£ 400.00	40	16000	Kitchen	Standard price	11	CA
29	16-APR-ES	Headphones	Bose	£ 230.00	20	4600	Electronics	Standard price	16	ES
30	07-MAR-CA	Dress	Zara	£ 60.00	30	1800	Fashion	Standard price	07	CA
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	400	Kitchen	Standard price	13	CA
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	650	Electronics	Standard price	24	CA
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	2500	Fashion	Standard price	02	CA
34	17-JUN-IN	Laptop	HP	£ 950.00	25	23750	Electronics	High Price	17	IN
35	09-JUL-FR	Watch	Casio	£ 100.00	20	2000	Accessories	Standard price	09	FR
36										
37			***Sales Report***							
38										
39		**Total Price of all Products**		£ 289,700.00						
40		**How many products that exists**		34						
41		**Average price of all products**		£ 297.06						
42										
43		**Minimum price of all Products**		£ 30.00						
44		**Maximum Price of all Products**		£ 1,000.00						
45										
46		**Total Price of all Products in (EC)**		£ 234,200.00						
47		**Number of Products price less than £100**		£ 11.00						
48										

Step 5: Text Formatting

- **LEFT FUNCTION**
- **RIGHT FUNCTION**
- **MIDDLE FUNCTION**

'Product ID' using the (LEFT) function.

Left Function:



I2 =LEFT(A2,2)

	C	D	E	F	G	H	I	J	K
1	ne	Price ($)	Quantity	Total Price	Category	Price Range	Date (Left)	Country Code (Rig	Month (MID)
2		£ 1,000.00	30	30000	Electronics	High Price	28	US	JAN
3		£ 80.00	15	1200	Fashion	Standard price	15	US	FEB
4		£ 130.00	40	5200	Kitchen	Standard price	03	US	MAR
5	g	£ 900.00	25	22500	Electronics	High Price	11	US	APR
6	e	£ 70.00	20	1400	Outdoor	Standard price	22	US	MAY
7		£ 200.00	45	9000	Electronics	Standard price	07	UK	JUN
8		£ 30.00	5	150	Fashion	Standard price	19	UK	JUL
9		£ 90.00	35	3150	Kitchen	Standard price	23	UK	AUG
10		£ 500.00	50	25000	Electronics	High Price	05	UK	SEP
11	d	£ 130.00	10	1300	Outdoor	Standard price	14	UK	OCT
12		£ 950.00	25	23750	Electronics	High Price	17	IN	JUN
13		£ 90.00	40	3600	Fashion	Standard price	25	AU	NOV
14	o	£ 120.00	35	4200	Kitchen	Standard price	08	DE	DEC
15		£ 150.00	15	2250	Electronics	Standard price	18	CA	FEB
16		£ 250.00	20	5000	Electronics	Standard price	16	ES	APR
17	e	£ 50.00	35	1750	Accessories	Standard price	21	CA	AUG
18		£ 160.00	15	2400	Electronics	Standard price	20	CN	AUG
19		£ 980.00	10	9800	Electronics	High Price	27	IT	JAN
20		£ 150.00	15	2250	Fashion	Standard price	01	UK	MAR
21	h	£ 200.00	10	2000	Outdoor	Standard price	14	US	AUG
22		£ 700.00	50	35000	Electronics	High Price	14	RU	MAY
23	c	£ 80.00	20	1600	Kitchen	Standard price	09	CA	JAN
24		£ 150.00	30	4500	Electronics	Standard price	19	BR	JUL
25	e	£ 50.00	35	1750	Accessories	Standard price	21	CA	AUG
26		£ 800.00	45	36000	Electronics	High Price	29	CA	SEP
27	h	£ 130.00	25	3250	Fashion	Standard price	03	CA	JUN
28		£ 400.00	40	16000	Kitchen	Standard price	11	CA	JUL
29		£ 230.00	20	4600	Electronics	Standard price	16	ES	APR
30		£ 60.00	30	1800	Fashion	Standard price	07	CA	MAR
31	h	£ 40.00	10	400	Kitchen	Standard price	13	CA	APR
32		£ 130.00	5	650	Electronics	Standard price	24	CA	MAY
33		£ 50.00	50	2500	Fashion	Standard price	02	CA	DEC
34		£ 950.00	25	23750	Electronics	High Price	17	IN	JUN
35		£ 100.00	20	2000	Accessories	Standard price	09	FR	JUL
36									

Dataset

The 'Product ID' column using the (RIGHT) function.

Right Function:

J2 =RIGHT(A2,2)

	C	D	E	F	G	H	I	J	K
1	ne	Price ($)	Quantity	Total Price	Category	Price Range	Date (Left)	Country Code (Rig	Month (MID)
2		£ 1,000.00	30	30000	Electronics	High Price	28	US	JAN
3		£ 80.00	15	1200	Fashion	Standard price	15	US	FEB
4		£ 130.00	40	5200	Kitchen	Standard price	03	US	MAR
5	g	£ 900.00	25	22500	Electronics	High Price	11	US	APR
6	e	£ 70.00	20	1400	Outdoor	Standard price	22	US	MAY
7		£ 200.00	45	9000	Electronics	Standard price	07	UK	JUN
8		£ 30.00	5	150	Fashion	Standard price	19	UK	JUL
9		£ 90.00	35	3150	Kitchen	Standard price	23	UK	AUG
10		£ 500.00	50	25000	Electronics	High Price	05	UK	SEP
11	d	£ 130.00	10	1300	Outdoor	Standard price	14	UK	OCT
12		£ 950.00	25	23750	Electronics	High Price	17	IN	JUN
13		£ 90.00	40	3600	Fashion	Standard price	25	AU	NOV
14	o	£ 120.00	35	4200	Kitchen	Standard price	08	DE	DEC
15		£ 150.00	15	2250	Electronics	Standard price	18	CA	FEB
16		£ 250.00	20	5000	Electronics	Standard price	16	ES	APR
17	e	£ 50.00	35	1750	Accessories	Standard price	21	CA	AUG
18		£ 160.00	15	2400	Electronics	Standard price	20	CN	AUG
19		£ 980.00	10	9800	Electronics	High Price	27	IT	JAN
20		£ 150.00	15	2250	Fashion	Standard price	01	UK	MAR
21	n	£ 200.00	10	2000	Outdoor	Standard price	14	US	AUG
22		£ 700.00	50	35000	Electronics	High Price	14	RU	MAY
23	c	£ 80.00	20	1600	Kitchen	Standard price	09	CA	JAN
24		£ 150.00	30	4500	Electronics	Standard price	19	BR	JUL
25	e	£ 50.00	35	1750	Accessories	Standard price	21	CA	AUG
26		£ 800.00	45	36000	Electronics	High Price	29	CA	SEP
27	n	£ 130.00	25	3250	Fashion	Standard price	03	CA	JUN
28		£ 400.00	40	16000	Kitchen	Standard price	11	CA	JUL
29		£ 230.00	20	4600	Electronics	Standard price	16	ES	APR
30		£ 60.00	30	1800	Fashion	Standard price	07	CA	MAR
31	n	£ 40.00	10	400	Kitchen	Standard price	13	CA	APR
32		£ 130.00	5	650	Electronics	Standard price	24	CA	MAY
33		£ 50.00	50	2500	Fashion	Standard price	02	CA	DEC
34		£ 950.00	25	23750	Electronics	High Price	17	IN	JUN
35		£ 100.00	20	2000	Accessories	Standard price	09	FR	JUL
36									

Dataset

The 'Product ID' column using the(MID) function.

Middle Function:

K2 =MID(A2,4,3)

	C	D	E	F	G	H	I	J	K
1	me	Price ($)	Quantity	Total Price	Category	Price Range	Date (Left)	Country Code (Rig	Month (MID)
2		£ 1,000.00	30	30000	Electronics	High Price	28	US	JAN
3		£ 80.00	15	1200	Fashion	Standard price	15	US	FEB
4		£ 130.00	40	5200	Kitchen	Standard price	03	US	MAR
5	g	£ 900.00	25	22500	Electronics	High Price	11	US	APR
6	e	£ 70.00	20	1400	Outdoor	Standard price	22	US	MAY
7		£ 200.00	45	9000	Electronics	Standard price	07	UK	JUN
8		£ 30.00	5	150	Fashion	Standard price	19	UK	JUL
9		£ 90.00	35	3150	Kitchen	Standard price	23	UK	AUG
10		£ 500.00	50	25000	Electronics	High Price	05	UK	SEP
11	id	£ 130.00	10	1300	Outdoor	Standard price	14	UK	OCT
12		£ 950.00	25	23750	Electronics	High Price	17	IN	JUN
13		£ 90.00	40	3600	Fashion	Standard price	25	AU	NOV
14	o	£ 120.00	35	4200	Kitchen	Standard price	08	DE	DEC
15		£ 150.00	15	2250	Electronics	Standard price	18	CA	FEB
16		£ 250.00	20	5000	Electronics	Standard price	16	ES	APR
17	e	£ 50.00	35	1750	Accessories	Standard price	21	CA	AUG
18		£ 160.00	15	2400	Electronics	Standard price	20	CN	AUG
19		£ 980.00	10	9800	Electronics	High Price	27	IT	JAN
20		£ 150.00	15	2250	Fashion	Standard price	01	UK	MAR
21	h	£ 200.00	10	2000	Outdoor	Standard price	14	US	AUG
22		£ 700.00	50	35000	Electronics	High Price	14	RU	MAY
23	c	£ 80.00	20	1600	Kitchen	Standard price	09	CA	JAN
24		£ 150.00	30	4500	Electronics	Standard price	19	BR	JUL
25	e	£ 50.00	35	1750	Accessories	Standard price	21	CA	AUG
26		£ 800.00	45	36000	Electronics	High Price	29	CA	SEP
27	h	£ 130.00	25	3250	Fashion	Standard price	03	CA	JUN
28		£ 400.00	40	16000	Kitchen	Standard price	11	CA	JUL
29		£ 230.00	20	4600	Electronics	Standard price	16	ES	APR
30		£ 60.00	30	1800	Fashion	Standard price	07	CA	MAR
31	h	£ 40.00	10	400	Kitchen	Standard price	13	CA	APR
32		£ 130.00	5	650	Electronics	Standard price	24	CA	MAY
33		£ 50.00	50	2500	Fashion	Standard price	02	CA	DEC
34		£ 950.00	25	23750	Electronics	High Price	17	IN	JUN
35		£ 100.00	20	2000	Accessories	Standard price	09	FR	JUL
36									

Dataset